

MA.6.NSO.3.4

Benchmark

MA.6.NSO.3.4 Express composite whole numbers as a product of prime factors with natural number exponents.

Benchmark Clarifications:

Clarification 1: Within this benchmark, expectations include using natural number exponents up to 5.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.3.1, MA.6.NSO.3.2, MA.6.NSO.3.3

Terms from the K-12 Glossary

- Base (of an exponent)
- Composite Number
- Exponent
- Factors (of positive whole numbers)
- Natural Number
- Prime Factorization
- Prime Number
- Whole Number

Vertical Alignment

Previous Benchmarks

- MA.4.AR.3.1
- MA.5.NSO.2.1, MA.5.NSO.2.2

Next Benchmarks

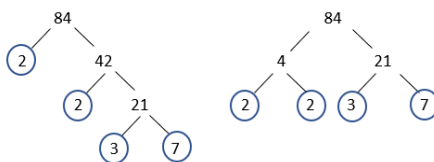
- MA.7.NSO.1.1

Purpose and Instructional Strategies

In grade 4, students determined factor pairs for whole numbers from 0 to 144 and determined if numbers are prime, composite, or neither. In grade 5, students multiplied and divided using products and divisors greater than 144. In future grades, students will use their understanding of factors and factorization with simplifying exponents, discovering exponent rules and generating equivalent rational expressions.

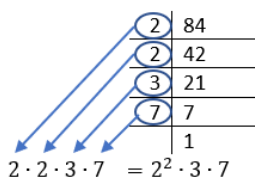
- Instruction includes representing multiplication in various ways.
 - $2^3 \times 3$
 - $2^3 \cdot 3$
 - $(2^3)(3)$
 - $2^3(3)$
- If the prime factorization has factors that occur more than once, it should be simplified where each factor is a unique term, and exponents are used to denote a prime factor occurring more than once (*MTR.2.1*).
- If a composite number has more than three factors, the factor tree can be started in more than one way. Students should identify and compare factor trees to other classmates to see that regardless of which factors they used for the first branch pair on the factor tree, it will result in the same prime factorization of the number.
- Instruction includes a variety of methods and strategies to determine the prime factorization (*MTR.2.1, MTR.4.1, MTR.5.1*).

- Factor tree



$$2 \cdot 2 \cdot 3 \cdot 7 = 2^2 \cdot 3 \cdot 7$$

- Successively dividing a number by prime numbers



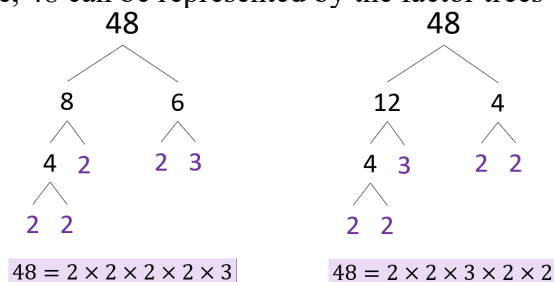
- The benchmark can be taught before 6.NSO.3.3 as a bridge between 6.NSO.3.1 and 6.NSO.3.3 and exponents.
- There is no limitation on the size of the whole number exponent for this benchmark.
- It is important to allow students to connect to the commutative property to understand that the prime factorization can be represented as $2^6 \cdot 3^3$ or $3^3 \cdot 2^6$ (*MTR.2.1, MTR.5.1*).

Common Misconceptions or Errors

- Students may incorrectly think when creating branches on a factor tree, one of the branches of each factor pair has to have a prime number.
- Students may incorrectly think the prime factorization of a number will change if different factors are selected for constructing the prime factorization model.
- Students may incorrectly write the prime factorization from a factor tree because they do not take all of the factors into their solution.
- Students may incorrectly think 2 is a composite number because it is even. Have students list the factor pairs that produce a product of 2 for them to conclude that 2 must be prime because it has exactly 2 factors: 1 and itself.
- Students may incorrectly think most odd numbers, like 91, are prime. It can be helpful to, within problem contexts, incorporate divisibility rules, especially those for 2, 3, 5, 6, 9 and 10, to help students identify factors quickly. If students believe a number is prime, students should try to find a number that can produce the one's digit and try numbers that end with that digit.
 - For example, 91 ends in a 1. If you multiply 7 by 3 you get a number that ends in a 1, and when the division is done, 7 is a factor of 91 with 13 being the second factor.
- Students may confuse finding factors with finding multiples of a number. It is important to use the academic vocabulary of factor and multiple within the contexts of problems and tasks.

Strategies to Support Tiered Instruction

- Teacher identifies the first seven prime numbers as 2, 3, 5, 7, 11, 13, 17 and introduces the divisibility rules to utilize when finding prime factors of a given value.
- Instruction includes comparing two different factor trees for the same given number with a focus on recognizing that multiple different factor pairs can be used to produce the same final factorization.
 - For example, 48 can be represented by the factor trees below.



- Instruction includes the use of additional divisibility rules for 2, 3, and 5 to assist with determining factors of a given value.
- Teacher creates and posts an anchor chart with visual representations of factors and multiples and encourages students to utilize the anchor chart to assist in utilizing correct academic vocabulary when referring to factors and multiples.
- Teacher provides students with flashcards to practice and reinforce academic vocabulary.
- Teacher models how to find a number that can produce the one's digit and try numbers that end with that digit.
 - For example, 91 ends in a 1. If you multiply 7 by 3 you get a number that ends in a 1, and when the division is done, 7 is a factor of 91 with 13 being the second factor. Teacher has students list the factor pairs that produce a product of 2 for them to conclude that 2 must be prime because it has exactly 2 factors: 1 and itself.

Instructional Tasks

Instructional Task 1 (MTR.2.1, MTR.3.1, MTR.4.1)

Lizzie and Adam both factored 864. Their answers are shown below:

Lizzie

$$864 = 2^5 \cdot 3^3$$

Adam

$$864 = 2 \cdot 2 \cdot 3 \cdot 3 \cdot 2 \cdot 2 \cdot 3 \cdot 2$$

Who factored 864 as a product of primes correctly? If Lizzie or Adam are incorrect, describe the error they made. If they are both correct, describe which factored form is more efficient as a solution and justify your reasoning.

Instructional Items

Instructional Item 1

Determine all of the factors of 24. Rewrite 24 as a product of its prime factors using exponents.

Instructional Item 2

Determine all of the factors of 216. Rewrite 216 as a product of its prime factors using exponents.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*